Grade 9 Physical Sciences – Study Notes

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The Physical Sciences paper is 1½ hours for 75 marks, covering Electricity, Forces, Magnetism and Electrostatics (Terms 3 and 4). Also learn the definitions on your class definition page.

Formula sheet

Learn these formulas, symbols and units. Every calculation in the paper uses one of them.

Quantity	Symbol	Unit (symbol)	Formula	Rearranged
Charge	Q	coulomb (C)	Q = I × t	I = Q ÷ t; t = Q ÷ I
Current	I	ampere (A)	I = Q ÷ t	
Potential difference	V	volt (V)	V = I × R	I = V ÷ R; R = V ÷ I
Resistance	R	ohm (Ω)	R = V ÷ I	
Energy delivered by a cell	W (or E)	joule (J)	W = V × Q	V = W ÷ Q; Q = W ÷ V
Time	t	second (s)		
Weight	w (or F_g)	newton (N)	w = m × g	m = w ÷ g
Mass	m	kilogram (kg)		
Gravitational field strength	g	newton per kilogram (N/kg)	g = 10 N/kg on Earth	
Electrical energy used	E	kilowatt-hour (kWh)	E = P (kW) × t (h)	
Cost	—	rand (R)	cost = kWh × price per kWh	

Triangle method

For a formula of the form A = B × C, write A at the top of a triangle and B and C at the bottom. Cover the quantity you want: if the other two are side by side, multiply; if one is above the other, divide.

- V = I × R → V on top, I and R at the bottom.
- Q = I × t → Q on top, I and t at the bottom.
- W = V × Q → W on top, V and Q at the bottom.
- w = m × g → w on top, m and g at the bottom.

Unit conversions

From	To	How
minutes	seconds	× 60
hours	seconds	× 3 600
milliampere (mA)	ampere (A)	÷ 1 000
gram (g)	kilogram (kg)	÷ 1 000
watt (W)	kilowatt (kW)	÷ 1 000
kilojoule (kJ)	joule (J)	× 1 000

How to set out a calculation (full marks)

1. Write the **formula**.
2. **Substitute** the values (in the correct units).
3. Write the **answer with the unit**.

Example: A current of 0.5 A flows for 2 minutes. Calculate the charge.

Q = I × t
Q = 0.5 × 120
Q = **60 C**

Marks are lost for a missing unit, the wrong unit, or forgetting to convert minutes to seconds.

Note: these notes use g = 10 N/kg. If your teacher uses 9.8 N/kg, use that value instead.

Electricity

Cells, batteries and energy conversion

A cell is a store of **chemical potential energy** that it changes into **electrical energy** when it is connected in a closed circuit.

Cells

- A cell has two **terminals**: a positive terminal (+) and a negative terminal (−).
- Inside is an **electrolyte** (a paste or liquid) and chemicals that react to push charges around the circuit.
- The chemical reaction only happens when the circuit is **closed**. That is why a cell lasts longer when the switch is off.
- When the chemicals have reacted completely, the cell can no longer supply energy: it is "flat".
- The **potential difference** (voltage) of a cell tells you how much energy it gives to each coulomb of charge. A 1.5 V cell gives 1.5 J to every coulomb that passes through it.

Types of cells:

Type	Description	Examples
Dry cell	Electrolyte is a paste; not rechargeable	AA, AAA torch batteries (1.5 V)
Wet cell	Electrolyte is a liquid	Car battery (lead–acid)
Rechargeable cell	Reaction can be reversed by charging	Phone and laptop batteries (lithium-ion)
Solar cell	Converts light energy directly into electrical energy	Solar panels, calculators

Batteries

A **battery** is **two or more cells connected together**.

- Cells connected **in series** (+ to −): the voltages **add up**. Three 1.5 V cells in series = 4.5 V.
- A bigger voltage pushes a bigger current through the same circuit, so bulbs glow brighter.

- In everyday language people call a single cell a "battery", but in science a battery has more than one cell.

Energy conversion in a cell and a circuit

Where	Energy conversion
Cell	Chemical potential energy → electrical energy
Light bulb	Electrical → light + heat (thermal)
Heater, kettle, iron	Electrical → heat
Electric motor, fan	Electrical → kinetic (movement)
Buzzer, speaker	Electrical → sound
Charging a phone battery	Electrical → chemical potential energy
Solar cell	Light (radiant) → electrical

Energy is **never created or destroyed**, only changed from one form to another (law of conservation of energy). Some energy is always "wasted" as heat — that is why a bulb and a cell get warm.

Calculating the energy delivered by a cell

W = V × Q (energy in J = potential difference in V × charge in C)

Example 1: A 3 V battery moves 200 C of charge through a circuit. How much energy does it deliver?

W = V × Q
W = 3 × 200
W = **600 J**

Example 2: A 1.5 V cell delivers a current of 0.2 A for 5 minutes. Calculate the energy.

Step 1 — charge: Q = I × t = 0.2 × 300 = 60 C
Step 2 — energy: W = V × Q = 1.5 × 60 = **90 J**

Example 3: A cell delivers 450 J of energy while 150 C passes through it. What is its voltage?

V = W ÷ Q = 450 ÷ 150 = **3 V**

Generating electricity, the national grid and cost

Most power stations work the same way: an energy source turns a **turbine**, the turbine turns a **generator**, and the generator produces electricity. A generator contains **magnets and coils of copper wire**; when the coil and magnet move relative to each other, a current is produced.

Methods of generating electricity

1. Coal-fired power stations (most of South Africa's electricity, run by Eskom, mainly in Mpumalanga)

- Coal is burnt → heats water → **steam** → steam turns the turbine → generator.
- Energy conversion: **chemical → heat (thermal) → kinetic → electrical**.
- Advantages: South Africa has lots of coal; reliable, works day and night.
- Disadvantages: **non-renewable**; air pollution (smoke, acid rain); releases **CO_2** (global warming); uses lots of water; mining damages land.

2. Nuclear power (Koeberg near Cape Town)

- Nuclear fission of uranium releases heat → steam → turbine → generator.
- Energy conversion: **nuclear → heat → kinetic → electrical**.
- Advantages: no CO_2; lots of energy from a small amount of fuel.
- Disadvantages: dangerous **radioactive waste**; risk of accidents; expensive to build.

3. Hydro-electric power

- Water stored behind a dam falls through pipes and turns turbines.
- Energy conversion: **gravitational potential → kinetic → electrical**.
- Advantages: **renewable**, no pollution. Disadvantages: needs big rivers and rain; dams flood land; SA is a dry country.

4. Wind power

- Wind turns the blades of a wind turbine.
- Energy conversion: **kinetic (wind) → electrical**.
- Advantages: renewable, clean. Disadvantages: only works when the wind blows; noisy; can harm birds.

5. Solar power

- **Solar (photovoltaic) panels** change sunlight directly into electricity (no turbine).
- Energy conversion: **light (radiant) → electrical**.
- Advantages: renewable, clean, lots of sunshine in SA. Disadvantages: only works in daylight; batteries needed for night; panels are expensive.

6. Gas / diesel (open-cycle gas turbines) — burnt fuel turns turbines; used at **peak times** to prevent load shedding; very expensive.

Renewable (will not run out)	Non-renewable (will run out)
Solar, wind, hydro, wave, geothermal	Coal, oil, natural gas, nuclear (uranium)

Provision of electricity and the national grid

The **national grid** is the network of power lines, transformers and substations that carries electricity from power stations to users all over the country.

1. **Power station** generates electricity (about 20 000 V).
2. **Step-up transformer** increases the voltage to about **400 000 V**.
3. **Transmission lines** on tall pylons carry the electricity over long distances. At high voltage the current is small, so **less energy is lost as heat** in the wires.
4. **Substations** with **step-down transformers** lower the voltage in stages.
5. **Distribution lines** deliver it to factories, schools and homes at **230 V**.

Load shedding happens when demand for electricity is higher than what the power stations can supply. Eskom switches off areas in turns so the whole grid does not collapse.

Cost of electricity

- **Power (P)** is how fast an appliance uses energy, measured in **watts (W)** or **kilowatts (kW)**. 1 kW = 1 000 W.
- Electricity is sold in **kilowatt-hours (kWh)** — one "unit" of electricity = using 1 kW for 1 hour.
- **Energy (kWh) = power (kW) × time (h)**
- **Cost = energy (kWh) × price per kWh**

Example 1: A 2 000 W heater is used for 3 hours. Electricity costs R2.50 per kWh.

P = 2 000 W ÷ 1 000 = 2 kW
Energy = 2 kW × 3 h = 6 kWh
Cost = 6 × R2.50 = **R15.00**

Example 2: A 100 W light bulb is left on for 10 hours a day for 30 days at R3 per kWh.

P = 0.1 kW; time = 10 × 30 = 300 h
Energy = 0.1 × 300 = 30 kWh
Cost = 30 × R3 = **R90.00**

Ways to save electricity: switch off lights and appliances (not on standby); use energy-saving **LED** bulbs; install a **geyser blanket** and turn the geyser down; use a **solar water heater**; boil only the water you need; dry clothes outside; use appliances with a good energy rating.

Circuit components and key terms

An electric circuit is a **closed path** through which charge can flow. It needs an energy source (cell), conducting wires, and something that uses the energy (resistor, bulb). If there is a break anywhere (open switch, broken wire), it is an **open circuit** and no current flows.

Circuit components

Component	Function	Circuit symbol
Cell	Provides energy to push charge around the circuit	A long thin line (+) and a short thick line (−), side by side
Battery	Two or more cells together; bigger voltage	Two or more cell symbols joined
Open switch	Breaks the circuit — no current	Line with a gap and a raised arm
Closed switch	Completes the circuit — current flows	Line with the arm closed
Light bulb	Changes electrical energy to light and heat	Circle with a cross (X) inside
Resistor	Opposes (limits) the current	Small rectangle
Variable resistor / rheostat	Resistance can be changed to control current (e.g. dimmer switch, volume knob)	Rectangle with a diagonal arrow through it
Ammeter	Measures **current**; always connected **in series**	Circle with **A**
Voltmeter	Measures **potential difference**; always connected **in parallel** across a component	Circle with **V**
Fuse	Thin wire that melts if the current is too high, breaking the circuit (safety)	Rectangle with a line through its length

Component	Function	Circuit symbol
Conducting wires	Connect the components	Straight lines (draw with a ruler, square corners)

Drawing circuit diagrams: use a ruler, straight lines and square corners; use the correct symbols; no gaps in a closed circuit.

Key terms

Charge (Q)

- Charge is carried by **electrons** moving through the conductor. Measured in **coulomb (C)**.
- Charge is not used up in a circuit — it flows around and back to the cell. It is the **energy** that is used up.

Current (I)

- Current is the **rate of flow of charge**: the amount of charge passing a point in the circuit **per second**.
- Unit: **ampere (A)**. 1 A = 1 coulomb per second.
- Measured with an **ammeter connected in series**.
- **Conventional current** flows from the **positive terminal to the negative terminal** outside the cell. Electrons actually flow the opposite way (− to +).

Potential difference (V)

- Potential difference is the **energy transferred per coulomb of charge** between two points in a circuit (also called **voltage**).
- Unit: **volt (V)**. 1 V = 1 joule per coulomb.
- Measured with a **voltmeter connected in parallel** across the component.
- The potential difference across the cell's terminals is the energy the cell gives to each coulomb.

Resistance (R)

- Resistance is the **opposition to the flow of current** in a conductor.
- Unit: **ohm (Ω)**.
- The bigger the resistance, the **smaller the current** for the same voltage.
- Resistance changes electrical energy into **heat** (and light in a bulb).

Calculations with charge and current: $Q = I \times t$

Example 1: A current of 3 A flows for 20 s. $Q = I \times t = 3 \times 20 =$ **60 C**.

Example 2: 120 C of charge passes a point in 1 minute. What is the current?

t = 60 s
$I = Q \div t = 120 \div 60 =$ **2 A**

Example 3: How long does it take for 90 C to pass if the current is 0.5 A?

$t = Q \div I = 90 \div 0.5 =$ **180 s** (= 3 minutes)

Calculations with V, I and R: $V = I \times R$

Example 1 (find V): A current of 0.2 A flows through a 30 Ω resistor.

$V = I \times R = 0.2 \times 30 =$ **6 V**

Example 2 (find I): A 12 V battery is connected to a 4 Ω resistor.

$I = V \div R = 12 \div 4 =$ **3 A**

Example 3 (find R): The voltmeter across a bulb reads 9 V and the ammeter reads 1.5 A.

$R = V \div I = 9 \div 1.5 =$ **6 Ω**

Series and parallel circuits

In a **series** circuit there is only **one path**, so the current is the same everywhere and the voltage is shared. In a **parallel** circuit there are **several paths (branches)**, so the voltage is the same across each branch and the current is shared.

Series circuits: theory

Components are connected **one after the other** in a single loop.

1. **Current is the same everywhere:** $I = I_1 = I_2 = I_3$. An ammeter reads the same value anywhere in the circuit.
2. **Potential difference is shared (divided):** $V = V_1 + V_2 + V_3$. The voltages across the components add up to the voltage of the battery.
3. **Total resistance adds up:** $R = R_1 + R_2 + R_3$.
4. Adding **more resistors (or bulbs) in series** → total resistance **increases** → current **decreases** → bulbs get **dimmer**.
5. If one bulb blows or a wire breaks, the circuit is open and **everything goes off**.
6. Adding more **cells** in series increases the voltage → bigger current → brighter bulbs.

Series circuits: calculations

Example 1: A 12 V battery is connected in series with a 2 Ω and a 4 Ω resistor.

(a) Total resistance: $R = R_1 + R_2 = 2 + 4 =$ **6 Ω**
(b) Current: $I = V \div R = 12 \div 6 =$ **2 A** (the same through both resistors)
(c) Voltage across 2 Ω: $V_1 = I \times R_1 = 2 \times 2 =$ **4 V**
(d) Voltage across 4 Ω: $V_2 = I \times R_2 = 2 \times 4 =$ **8 V**
Check: 4 + 8 = 12 V ✓

Example 2: In a series circuit the battery gives 9 V. A voltmeter across bulb 1 reads 3 V. What does a voltmeter across bulb 2 read?

$V = V_1 + V_2 \rightarrow 9 = 3 + V_2 \rightarrow V_2 =$ **6 V**

Example 3: Ammeter A_1 before the first bulb reads 0.4 A. What does ammeter A_2 after the second bulb read?

In series the current is the same everywhere: **0.4 A**.

Parallel circuits: theory

Components are connected on **separate branches**, side by side, between the same two points.

1. **Potential difference is the same across each branch:** $V = V_1 = V_2$. Each branch gets the full battery voltage.
2. **Current splits** at a junction and joins again: $I = I_1 + I_2$. The total current from the battery equals the sum of the branch currents.
3. More current flows through the branch with the **smaller resistance**.
4. Adding **more resistors in parallel** → total resistance **decreases** → total current from the battery **increases**.
5. Bulbs in parallel stay **equally bright** when more are added (each still gets the full voltage), but the battery goes flat faster.
6. If one bulb blows, the **others stay on** — that is why houses are wired in parallel.
7. Total resistance of two resistors in parallel: $1/R = 1/R_1 + 1/R_2$, or $R = (R_1 \times R_2) \div (R_1 + R_2)$. It is always **smaller than the smallest** resistor.

Parallel circuits: calculations

Example 1: A 6 V battery is connected to a 3 Ω resistor and a 6 Ω resistor in parallel.

(a) Voltage across each resistor: **6 V** (same as the battery)
(b) Current through 3 Ω: $I_1 = V \div R_1 = 6 \div 3 =$ **2 A**
(c) Current through 6 Ω: $I_2 = V \div R_2 = 6 \div 6 =$ **1 A**

(d) Total current: $I = I_1 + I_2 = 2 + 1 =$ **3 A**
(e) Total resistance: $R = V \div I = 6 \div 3 =$ **2 Ω** (smaller than 3 Ω ✓)

Example 2: The ammeter next to the battery reads 1.2 A. The ammeter in branch 1 reads 0.5 A. What is the current in branch 2?

$I = I_1 + I_2 \rightarrow 1.2 = 0.5 + I_2 \rightarrow I_2 =$ **0.7 A**

Example 3: Two identical bulbs are in parallel across a 3 V cell. The total current is 0.6 A. What is the current through each bulb?

Identical bulbs share the current equally: $0.6 \div 2 =$ **0.3 A** each.

Series vs parallel summary

	Series	Parallel
Paths for current	One	More than one
Current	Same everywhere	Splits: $I = I_1 + I_2$
Potential difference	Shared: $V = V_1 + V_2$	Same: $V = V_1 = V_2$
Total resistance	$R = R_1 + R_2$ (increases)	Decreases as resistors are added
Add a bulb	Others get dimmer	Others stay the same brightness
One bulb breaks	All go out	Others stay on
Example	Old Christmas lights, torch cells	House wiring, car headlights

Factors that affect the resistance of a conductor

Resistance happens because moving electrons **collide with the atoms** of the conductor. Anything that causes more collisions increases resistance.

Factor	Effect	Why
Length	Longer wire → **more** resistance (double the length = double the resistance)	Electrons travel further and collide with more atoms
Thickness (cross-sectional area)	Thicker wire → **less** resistance	More space for electrons to flow, like a wider road

Factor	Effect	Why
Type of material	Silver and copper have **low** resistance; nichrome and tungsten have **high** resistance	Different materials hold on to their electrons differently
Temperature	Hotter wire → **more** resistance	Atoms vibrate faster, so electrons collide more

Conductors and insulators

- **Conductors** have low resistance and let current flow easily: metals such as copper, aluminium, silver, gold, iron; also graphite and salt water.
- **Insulators** have very high resistance and do not let current flow: plastic, rubber, wood, glass, dry air.
- Electrical wires are made of **copper** (good conductor) covered in **plastic** (insulator for safety).
- **Nichrome** wire (high resistance) is used in heating elements of kettles, toasters and heaters. **Tungsten** is used in bulb filaments because it can get very hot without melting.

Investigating resistance (practical)

- **Aim:** to find how the length of a nichrome wire affects its resistance.
- **Independent variable:** length of the wire (what you change).
- **Dependent variable:** current / resistance (what you measure).
- **Controlled variables:** thickness and type of wire, temperature, the battery (keep the same).
- **Result:** as the length increases, the ammeter reading **decreases**, so the resistance **increases**.
- **Conclusion:** the longer the wire, the greater its resistance.

Forces

What a force is, its effects and types

A **force (F)** is a **push or a pull** that one object exerts on another. Forces always act **between two objects**, and a force always has a **size (magnitude)** and a **direction** — so we draw forces as arrows.

- Unit of force: **newton (N)**, named after Sir Isaac Newton.

- 1 N is roughly the weight of a small apple (about 100 g).
- Forces are measured with a **spring balance** (newton meter).

4 effects of forces

Effect	Example
1. **Change the shape** of an object	Squashing a ball, stretching a spring, bending a ruler
2. **Start or stop** an object moving	Kicking a stationary ball; a goalkeeper catching it
3. **Change the speed** of an object (speed up or slow down)	Pushing a swing harder; brakes slowing a car
4. **Change the direction** of motion	A batsman hitting a cricket ball; steering a bicycle

If all the forces on an object are **balanced** (cancel out), the object stays still or keeps moving at the same speed in a straight line. **Unbalanced** forces cause a change in motion.

Two categories of forces

- **Contact forces** act only when the objects are **touching**.
- **Field forces** (non-contact forces) act **at a distance**, without the objects touching. They act within a **field** (gravitational, electric or magnetic).

Contact forces

Applied force (F_A)

- A push or pull applied directly by a person or object.
- Examples: pushing a trolley, pulling a door, kicking a ball.

Friction (F_f)

- Force that **opposes motion** between two surfaces in contact. It always acts **opposite to the direction of motion** (or attempted motion), parallel to the surface.
- Rough surfaces → more friction; smooth surfaces → less friction.
- Friction produces **heat** (rub your hands together).
- Useful: grip of shoes and tyres, brakes, walking, writing with a pencil, striking a match.
- A nuisance: wears out shoes, tyres and machine parts; wastes energy as heat. Reduce it with oil, grease, ball bearings, smooth surfaces and streamlining.

- **Air resistance (drag)** is friction from air; **water resistance** from water.

Normal force (F_N)

- The force a surface exerts on an object resting on it. It acts **perpendicular (at 90°) to the surface**, pushing outwards.
- A book on a table: the table pushes **up** on the book with a normal force equal to the book's weight.

Tension (F_T)

- The pulling force in a stretched **rope, string, chain or cable**. It acts **along the rope, away from the object** it is attached to.
- Examples: a lamp hanging from a cable, a tug-of-war rope, a crane lifting a load, a dog pulling on a lead.

Field forces

Weight / gravitational force (w or F_g)

- The force with which the **Earth pulls an object towards its centre**. Always drawn **straight down**.
- Every object with mass attracts every other object (gravity), but the effect is only noticeable for very large masses such as planets.

Electrostatic force

- Force between **charged objects**. Like charges repel; unlike charges attract. (See Electrostatics.)

Magnetic force

- Force between **magnets**, or between a magnet and a **magnetic material** such as iron. Like poles repel; unlike poles attract. (See Magnetism.)

Contact forces	Field forces
Applied force	Weight (gravitational force)
Friction	Electrostatic force
Normal force	Magnetic force
Tension	

Force diagrams, free body diagrams, mass and weight

Force diagrams vs free body diagrams

	Force diagram	Free body diagram
How the object is drawn	As a picture or a box	As a **dot**
Where arrows start	On the object, where the force acts	From the **dot**
Use	Shows the situation	Shows only the forces acting **on** the object

Rules for drawing

1. Draw only forces acting **on** the object (not forces the object exerts on other things).
2. Each force is an **arrow** pointing in the direction of the force.
3. **Longer arrow = bigger force.** Equal forces get equal-length arrows.
4. **Label** every arrow with its name or symbol and size if given: w / F_g (weight), F_N (normal), F_A (applied), F_f (friction), F_T (tension).
5. Use a ruler.

Common situations

Situation	Forces to draw
Book resting on a table	Weight down; normal force up (same length)
Box pushed to the right at constant speed	Weight down; normal force up; applied force right; friction left (same length as applied force)
Box pushed to the right and speeding up	As above, but the **applied force arrow is longer** than friction
Lamp hanging from a cable	Weight down; tension up (same length)
Trolley pulled with a rope to the left	Weight down; normal up; tension left (along the rope); friction right
Ball falling through the air	Weight down; air resistance up (shorter, if it is speeding up)

Situation	Forces to draw
Magnet hanging above a paperclip lifting it	On the paperclip: weight down; magnetic force up

Mass and weight

Mass (m) is the **amount of matter** in an object. **Weight (w)** is the **gravitational force** of the Earth (or another planet) **on** the object.

	Mass	Weight
Definition	Amount of matter in an object	Gravitational force with which the Earth pulls on an object
Unit	kilogram (kg)	newton (N)
Instrument	Balance / scale	Spring balance (newton meter)
Is it a force?	No	Yes — a field force
Direction	None	Always downwards, towards the centre of the Earth
Same everywhere?	**Yes** — mass does not change on the Moon	**No** — depends on gravity; on the Moon you weigh about ⅙ of your Earth weight

w = m × g, where g is the **gravitational field strength**: about **10 N/kg** on Earth and **1.6 N/kg** on the Moon.

In everyday language people say "I weigh 50 kg", but scientifically that is your **mass**; your weight is 500 N.

Mass and weight calculations

Example 1 (find weight): Calculate the weight of a 45 kg learner on Earth.

w = m × g
w = 45 × 10
w = **450 N**

Example 2 (grams): Calculate the weight of a 250 g packet of sugar.

m = 250 ÷ 1 000 = 0.25 kg
w = 0.25 × 10 = **2.5 N**

Example 3 (find mass): A box has a weight of 800 N. What is its mass?

m = w ÷ g = 800 ÷ 10 = **80 kg**

Example 4 (Moon): An astronaut has a mass of 70 kg. Calculate the weight on Earth and on the Moon (g = 1.6 N/kg).

Earth: w = 70 × 10 = **700 N**
Moon: w = 70 × 1.6 = **112 N**
The mass stays **70 kg** in both places.

Magnetism

Magnetic force, poles and fields

A **magnet** is an object that attracts magnetic materials and has a magnetic field around it. The **magnetic force** is the force a magnet exerts on another magnet or a magnetic material; it is a **field force** (acts without touching).

Poles of magnets

- Every magnet has two poles: a **north-seeking pole (N)** and a **south-seeking pole (S)**.
- The magnetic force is **strongest at the poles**; in the middle of a bar magnet it is weakest.
- **Law of magnetism: like poles repel, unlike poles attract.** N–N and S–S push apart; N–S pull together.
- Poles always come in **pairs**. If you cut a bar magnet in half, you get two smaller magnets, each with its own N and S pole. You can never get a single pole.
- A freely suspended magnet (or compass needle) swings until its **N pole points to the Earth's geographic North**. That is how the poles got their names.
- **Repulsion is the only sure test for a magnet:** a magnet attracts both a magnet and unmagnetised iron, but it can only repel another magnet.
- The magnetic force gets **weaker with distance** and passes through non-magnetic materials such as paper, wood and glass.

Magnetic fields

A **magnetic field** is the **region around a magnet where a magnetic force is experienced** by a magnet or magnetic material.

You can show the field by sprinkling **iron filings** on paper over a magnet, or by moving a small **plotting compass** around the magnet and marking the direction of its needle.

Rules for drawing magnetic field lines:

1. Field lines go **from the N pole to the S pole** outside the magnet. Show the direction with arrows.
2. Field lines **never cross** each other.
3. Field lines are **closest together where the field is strongest** — at the poles.
4. Lines leave and enter the poles; draw them as smooth curves.

Magnetic field patterns of bar magnets

Arrangement	What the pattern looks like	Force
One bar magnet	Curved loops leave the N pole, curve around both sides, and enter the S pole. Lines are crowded at the poles and spread out at the sides.	—
Two magnets, unlike poles facing (N → S)	Lines run **straight across the gap** from the N pole to the S pole, close together and evenly spaced; the outer lines curve around.	**Attraction**
Two magnets, like poles facing (N–N or S–S)	Lines from each pole **bend away** from each other and do not connect. In the middle there is a **neutral point** where there is no field (the forces cancel). Arrows point away from N–N poles, towards S–S poles.	**Repulsion**

Magnetic substances and ferromagnetic materials

- **Magnetic materials** are attracted by a magnet and can be made into magnets.
- **Non-magnetic materials** are not attracted: wood, plastic, glass, paper, rubber, and most metals such as **copper, aluminium, gold, silver, brass and tin**.
- **Ferromagnetic materials** are **strongly attracted** to magnets and can be **magnetised**. There are three ferromagnetic elements: **iron (Fe), nickel (Ni) and cobalt (Co)**, plus their alloys such as **steel**.
- **Not all metals are magnetic** — a common exam trap. A cold-drink can (aluminium) is not attracted; a food can (steel) is.

Soft and hard magnetic materials:

	Soft iron	Steel
Magnetises	Easily	With more difficulty
Keeps its magnetism	No — loses it quickly (**temporary magnet**)	Yes (**permanent magnet**)
Used for	Electromagnets, transformer cores	Fridge magnets, compass needles, speakers

How to magnetise: stroke an iron or steel nail many times **in one direction** with one pole of a magnet; or place it in a coil of wire carrying a current (electromagnet). **How to demagnetise:** heat it strongly, hammer or drop it repeatedly.

Uses of magnets: compasses, electric motors and generators, speakers, fridge doors, MRI scanners, credit-card strips, sorting scrap iron from other metals, maglev trains.

The magnetic field of the Earth

- The Earth behaves like a **giant bar magnet**. Its magnetic field is caused by the movement of **molten iron and nickel in the outer core**.
- The **Earth's magnetic south pole is near the geographic North Pole**, and its magnetic north pole is near the geographic South Pole. The N pole of a compass is attracted to it (unlike poles attract), so the compass points north.
- The magnetic poles are **not in exactly the same place** as the geographic poles, and they **move slowly** over time.
- The Earth's magnetic field (the **magnetosphere**) **protects Earth from the solar wind** — harmful charged particles from the Sun — and so protects living things.
- Near the poles, some charged particles enter the atmosphere and make gases glow: the **aurora** (Aurora Borealis in the north, Aurora Australis in the south).
- **Uses:** navigation with a compass (hikers, ships, aeroplanes); birds, turtles and whales use the field to find their way when they migrate.

Electrostatics

Charges, the electrostatic force and conservation of charge

Electrostatics is the study of charges that are **not moving** (static electricity). Objects become charged when **electrons are transferred** from one object to another.

Charges in atoms

Particle	Charge	Where	Can it move between objects?
Proton	Positive (+)	In the nucleus	**No** — held tightly in the nucleus
Neutron	No charge (neutral)	In the nucleus	No
Electron	Negative (−)	Around the nucleus	**Yes** — the outer electrons can be transferred

- A **neutral** object has **equal numbers of protons and electrons**, so the charges cancel out. It still contains charges.
- Charge is measured in **coulomb (C)**. One electron has a very small charge, so 1 C is the charge of billions of electrons.

Positive and negative charges: how objects become charged

When two **different materials** (insulators) are **rubbed together**, friction causes **electrons to move from one material to the other**. This is called **charging by friction**.

The object that…	Has…	Its charge is
gains electrons	more electrons than protons	**negative (−)**
loses electrons	fewer electrons than protons	**positive (+)**

Only electrons move. An object becomes positive by **losing** electrons, never by gaining protons.

Examples:

Rubbed together	Gains electrons (−)	Loses electrons (+)
Plastic ruler and woollen cloth	Plastic ruler	Wool
Glass rod and silk cloth	Silk	Glass rod
Balloon and hair	Balloon	Hair
Ebonite rod and fur	Ebonite rod	Fur

The two objects always end up with **opposite charges of the same size**.

Conductors and insulators: insulators (plastic, glass, rubber) **keep** the charge where it was rubbed, so they are easy to charge. In conductors (metals) electrons move freely, so a metal held in your hand loses its charge to the ground through your body (it is **earthed**).

The electrostatic force

The **electrostatic force** is the force of **attraction or repulsion between charged objects**. It is a **field force** — it acts without the objects touching.

- **Like charges repel** (+ and +, − and −).
- **Unlike charges attract** (+ and −).
- A charged object **also attracts neutral objects**: a charged comb picks up small pieces of paper, and a charged balloon sticks to a wall. (The charge on the comb pulls the opposite charges in the paper slightly closer.)
- The force is **stronger when the objects are closer** together and when they carry **more charge**.

Demonstration: hang a charged plastic ruler on a string. Bring a second charged plastic ruler near it → they **repel** (both −). Bring a charged glass rod near it → they **attract** (+ and −).

Conservation of charge

The principle of conservation of charge: charge **cannot be created or destroyed**; it can only be **transferred** from one object to another. The **total charge of an isolated system stays the same**.

Example: A neutral plastic ruler and a neutral cloth are rubbed together. 20 electrons move from the cloth to the ruler.

- Before: ruler = 0, cloth = 0 → total = **0**
- After: ruler has 20 extra electrons (−), cloth is missing 20 electrons (+) → total = (−20) + (+20) = **0**
- The total charge is the same before and after: charge is **conserved**.

Static electricity in everyday life

Examples:

- A small shock when you touch a car door or a metal handle after walking on a carpet.
- Hair standing up after taking off a jersey; clothes sticking together after the tumble dryer.
- **Lightning:** charges build up in thunderclouds and discharge to the ground or another cloud as a huge spark.

Uses: photocopiers and laser printers; spray painting cars (charged paint droplets are attracted to the car body, less waste); dust precipitators in power-station chimneys that remove smoke particles.

Dangers and safety:

- Sparks can ignite petrol, gas or flour dust — fuel trucks are **earthed** with a chain or cable while filling.
- **Lightning conductors** (metal rods on tall buildings) carry the charge safely into the ground.
- During a thunderstorm, stay indoors; don't shelter under a tree or stand in open fields.